

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

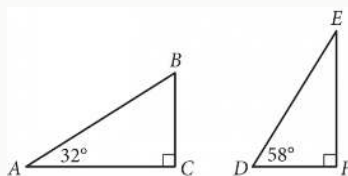
Geometry and Trigonometry

05

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

A.

$$\frac{DE}{DF}$$

B.

$$\frac{DF}{DE}$$

C.

$$\frac{DF}{EF}$$

D.

$$\frac{EF}{DE}$$

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

GRID-IN

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in degrees?

.	/	.	/	.	.
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

GEOMETRY AND TRIGONOMETRY — SET 5 · GEOMETRY AND TRIGONOMETRY

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

- A. 34
- B. 102
- C. 136
- D. 170

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

- A. 3
- B. 4
- C. 8
- D. 27

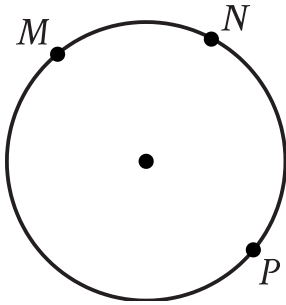
The length of the edge of the base of a right square prism is 6 units. The volume of the prism is 2,880 cubic units. What is the height, in units, of the prism?

A. $4\sqrt{30}$

B. 36

C. $24\sqrt{5}$

D. 80



- Clockwise from top left, the following points are on the circle:
 - upper M
 - upper N
 - upper P

Points M , N , and P lie on the circle shown. On this circle, minor arc MN has a length of 39 centimeters and major arc MPN has a length of 195 centimeters. What is the circumference, in centimeters, of the circle shown?

- A. 39
- B. 156
- C. 195
- D. 234

One leg of a right triangle has a length of 43.2 millimeters. The hypotenuse of the triangle has a length of 196.8 millimeters. What is the length of the other leg of the triangle, in millimeters?

- A. 43.2
- B. 120
- C. 192
- D. 201.5

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k ?

A. $\frac{29}{28}$

B. 1

C. $\frac{28}{29}$

D. 0

Circle K has a radius of 4 millimeters mm. Circle L has an area of 100π mm². What is the total area, in mm², of circles K and L ?

- A. 14π
- B. 28π
- C. 56π
- D. 116π